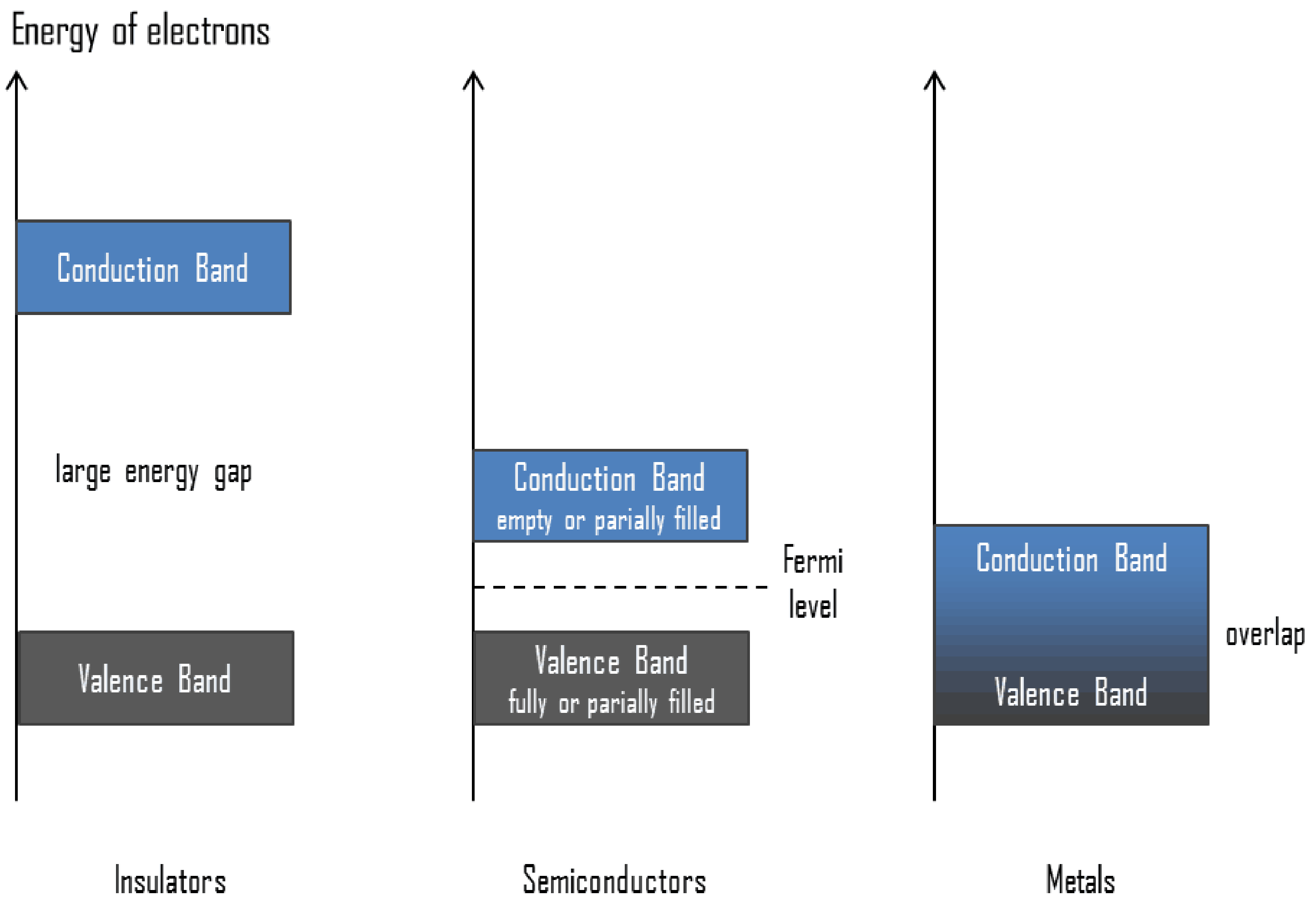
Semiconductors

- Fundamental of semiconductor
  - On the basis of electrical conductivity, materials are generally classified in to three types; conductors, insulators and semiconductors.

Material	Resitivity, $\rho(\Omega\text{-m})$	Temperature Coefficient, $\alpha(c^{o})^{-1}$
Conductors		
Silver	1.59×10^{-8}	0.0061
Copper	1.68×10^{-8}	0.0068
Gold	2.44×10^{-8}	0.0034
Aluminium	2.65×10^{-8}	0.00429
Tungsten	5.6×10^{-8}	0.0045
Iron	9.71×10^{-8}	0.00651
Platinum	10.6×10^{-8}	0.003927
Mercuy	98×10^{-8}	0.0009
Nichrome(Ni,Fe,Cr alloy)	100×10^{-8}	0.0004
Semiconductors		
Carbon(Graphite)	$(3\text{-}60)\times10^{-5}$	-0.0005
Germanium	$(1\text{-}500)\times10^{-3}$	-0.05
Silicon	0.1 - 60	-0.07
Insulators		
Glass	$10^{9}\text{-}10^{12}$	
Hard rubber	$10^{13}\text{-}10^{15}$	

- This behavior can be explained using band diagram.

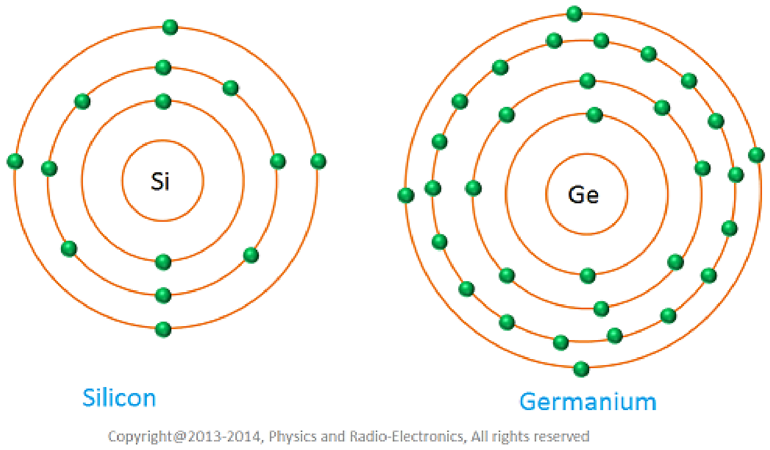


- Insulators
  - A solid is said to be an insulator, if no current exists when we apply a potential difference across it.
  - Insulators are materials in which valance electrons are bond very tightly to their parent atoms.
  - Required very large energy to remove them from the nuclei.
  - In terms of energy band structure, the conduction band and valance band on an insulator is separated by a very large forbidden energy gap (about 5.5 eV).
  - For these type of materials, the probability of valance electrons acquiring sufficient energy to overcome the energy gap is negligible at RT and applied electric field.
  - Therefore, virtually no effective electrons are present in the conduction band and current cannot flow in these materials.

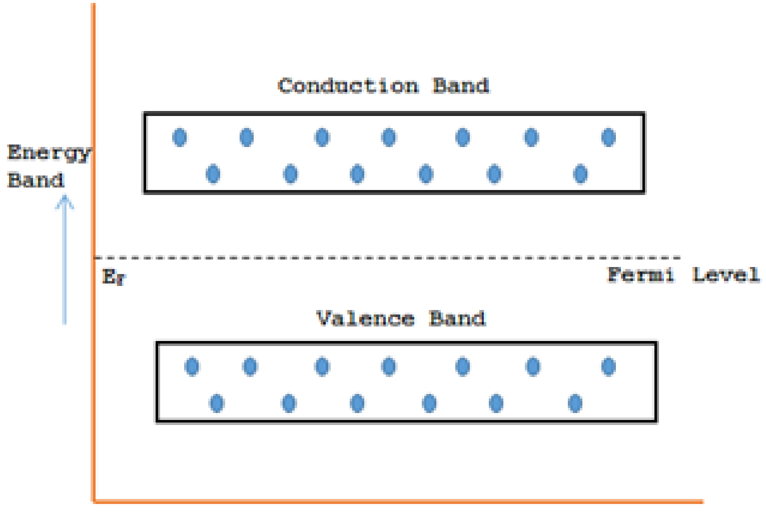
- Conductors
  - In terms of energy bands, the conduction band and valance band in the conductors are very close to one another or may be overlap.
  - Due to thermal vibrations, the conduction band is partially filled with valance electrons which moves randomly.
  - With applied electric field, electrons at the top of the valance band acquire sufficient energy to raise them into the conduction band and they also move with drift velocity in one direction.
  - These electrons are readily available for conduction as a flow of current through the conductor.

- Semiconductors
  - They are neither a very good conductor nor very good insulator.
  - Their conductivity can be increased by adding tiny but controlled amounts of other substances call “impurities”.
  - Semiconductors have a forbidden gap between filled valance band and unoccupied conduction band.
  - This gap is small compared to the gap of insulators.
  - The narrow energy gap in semiconductor (Si – 1.2 eV, Ge – 0.7 eV) imply that some electrons from the completely filled lower valance band can be thermally excited to the empty conduction band.
  - Excited electrons are able to conduct small current on application of electric field.
  - At the room temperature they have,
    - Partially filled conduction band.
    - Partially filled valance band.
    - Very narrow forbidden gap E_g (< 2eV).

- Semiconductor materials
  - Semiconductor play a vita role in modern electronic components, such as diodes, transistors, etc..
  - Semiconductor may be,
    - Element material – Germanium, Silicon.
    - Compound material – GaAs, Indium Phosphide.
    - Alloys – AlGaAs
  - Most frequently used semiconductor materials are Si, Ge.
  - Si has 14 electrons. Two in the K shell, 8 in the L shell and 4 in the M shell. Therefore, Si atom has 4 valance electrons.
  - Ge has an atomic number 32 and it also has 4 valance electrons. Both of these are tetravalent or group IV elements in the periodic table.



- Intrinsic (pure) semi-conductor



Intrinsic semiconductor - Energy and Diagram

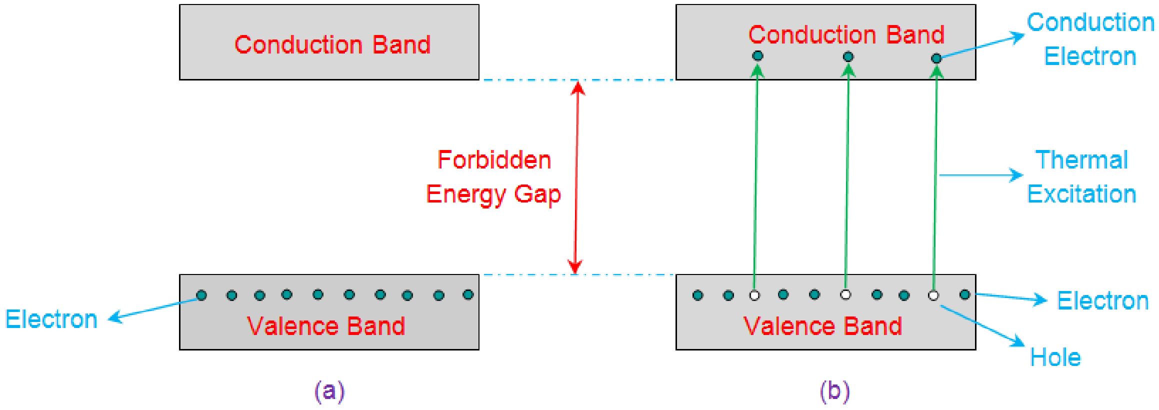


Figure 2 Energy Band Diagram of Intrinsic Semiconductor at (a) 0K (b) Temperature > 0K

- E_f is located at about middle of the gap and since E_g is small, appreciable number of electrons are thermally excited from VB to CB.
- A small applied voltage can easily raise the energy of the electrons in the CB (moderate current)
- Conductivity of a semi-conductor increases with temperature. (depends strongly on temperature)
- In intrinsic semiconductor, densities of free electrons in conduction band and free holes in valance band are equal. (i.e. $n = p = n_i \propto e^{-E_g/KT}$)

- Under the action of the electric field E established by the applied potential difference, the electrons acquire an average drift velocity v_e which is related to the applied electric field.

$$v_e = -\mu_e E$$

- Where the μ_e is known as electron mobility. The minus sign indicate that the direction of the drift velocity is opposite direction to the electric field.
- Similarly, the average drift velocity $v_h$for the holes given by

$$v_h = \mu_h E$$

- Then the overall conductivity σ of the pure semiconductor is given by,

$$\sigma = \sigma_e + \sigma_h = n_e e \mu_e + n_h e \mu_h$$

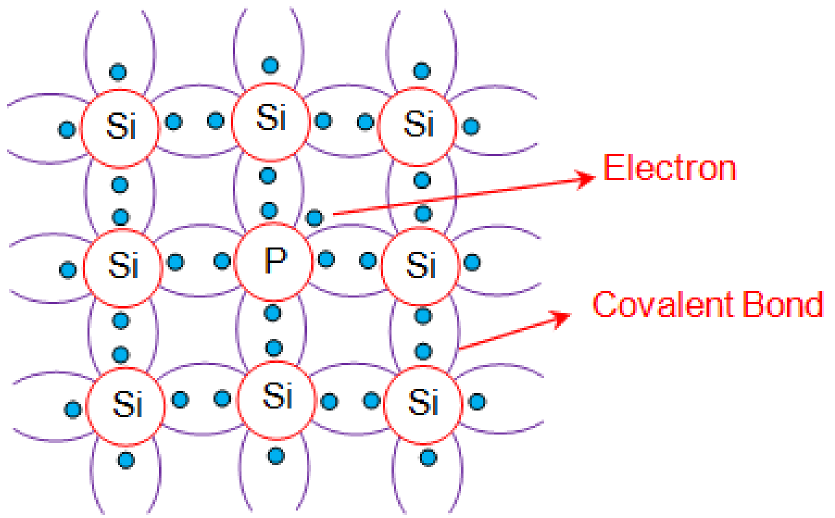
- Where σ_e and σ_h are the conductivity of semiconductor due to electron and holes respectively. e is the electronic charge. Hence, the resistivity *r* of pure semiconductor is given by

$$\rho = \frac{1}{\sigma} = \frac{1}{n_e e \mu_e + n_h e \mu_h}$$

- At a given temperature, electron carrier density n_e and the hole carrier density $n_h$of intrinsic semiconductor are equal ($n_i = n_e = n_h$).

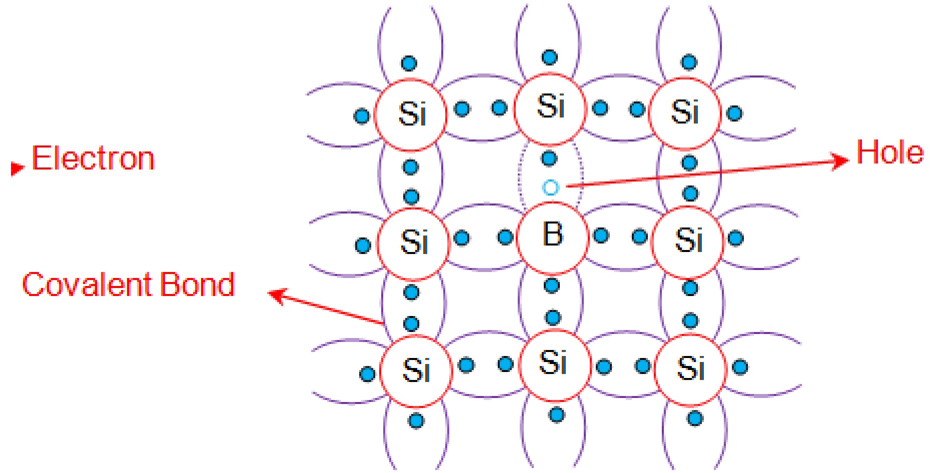
$$\sigma = n_i e(\mu_e + \mu_h) \text{ and } \rho = \frac{1}{n_i e(\mu_e+\mu_h)}$$

- Extrinsic semi-conductors
  - Extrinsic semi-conductors are doped semiconductors.
  - Normally to increase the conductivity of intrinsic semiconductors (small # electrons present at RT), impurities are added to them.
  - Ex – an atom with five valance electrons such as Arsenic (As) is added to a semiconductor. Then four valance electrons (doner impurities) participate in the covalent bond and one electron is left over.



  - The extra electron is nearly free and has an energy level within the energy gap just below conduction band.
  - So small amount of thermal energy will cause an electron to move into the conduction band.
  - Semiconductors doped with donor atoms are called n-type semiconductors.

- If a semiconductor is doped with three atoms valance electrons (In, Al), These three electrons from covalent bonds with neighboring atoms, leaving an electron deficiency or hole in the forth bond.



- The energy levels of such impurities lie with in the energy gap, just above the valance band.
- Such impurities are referred as acceptors.
- Semiconductors doped with acceptor impurities are known as p-type semiconductors.

- Doping
  - The pure semiconductors like Si or Ge, has an impurity of about one part in 10^{10}. However, if an impurity is added even as a small amount of one part in 10^7, the conductivity of the material can change considerably by about a factor 10^3.

- Conductivity of Extrinsic semiconductor
  - In p-type semi conductors, $n_h \gg n_e$, then conductivity is given by

$$\sigma_h = n_h e \mu_h$$

  - In n-type semi conductors, $n_h \ll n_e$, then conductivity is given by

$$\sigma_e = n_e e \mu_e$$

  - When the temperature is increased further from room temperature, the number of thermally generated electron-hole pair will increase as it happens in an intrinsic semiconductor. As the temperature keep rising, the electron-hole pair pairs will continuously keep increasing and minority carriers also increase. The process will lead the doped (extrinsic) semiconductor to behave like an intrinsic semiconductor.

- Carrier concentration in semiconductor
  - The carrier densities in an extrinsic semiconductor material depends on the concentration of the impurity atoms introduced by doping. Since the material must remain electrically neutral after doping, the number of positive and negative charges must be equal. If N_A and N_D are the concentrations of impurity atoms in acceptor and doner materials, which are assume to be fully ionized.
  - Then for n type semiconductor

$$n_e = n_h + N_D$$

  - For p type semi conductor

$$n_h = n_e + N_A$$

  - Where n_e and n_h are the electron carrier density and hole carrier density.
  - In an intrinsic semiconductor, we have seen $n_e = n_h = n_i$

$$n_e n_h = n_i^2$$

  - For n-type semiconductor, $N_D \gg n_i$

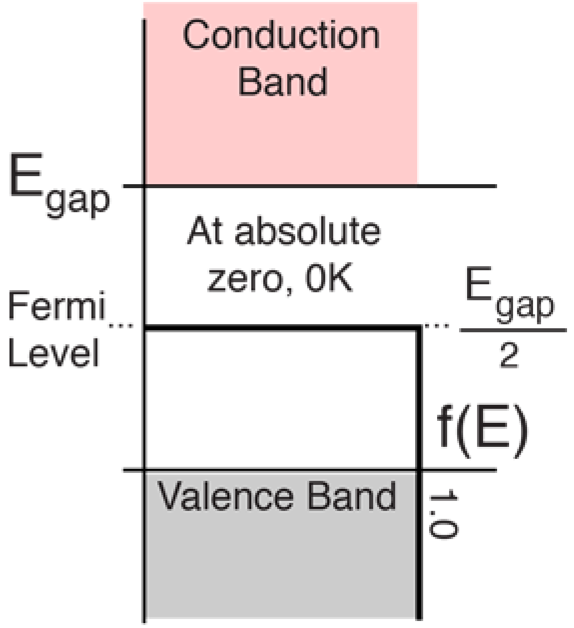
$$n_e \approx N_D \text{ and } n_h \approx {n_i^2}/{N_D}$$

  - For p-type semiconductor, $N_A \gg n_i$

$$n_h \approx N_A \text{ and } n_e \approx {n_i^2}/{N_A}$$

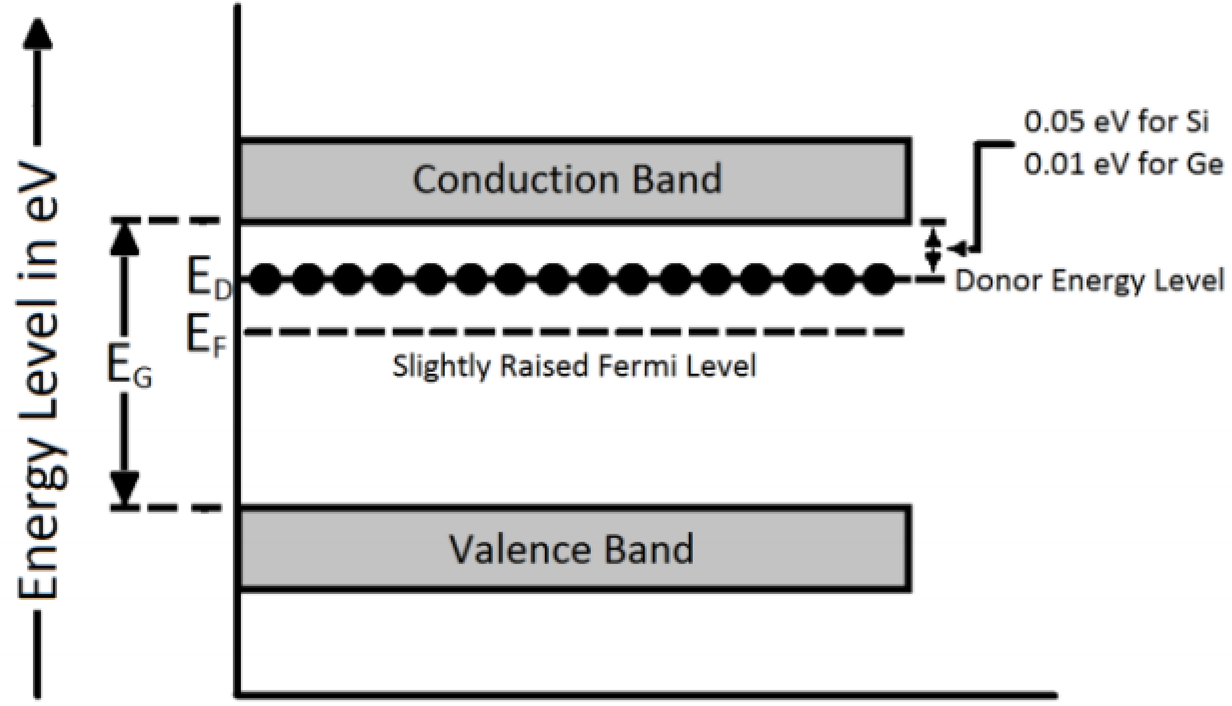
- Fermi level
  - Fermi level is a measure of the probability of finding the occupancy of allowed energy state.

- Fermi level of intrinsic semiconductors.
  - In case of intrinsic or pure semiconductors, at absolute zero (0K), all atomic electrons would be bound to the crystal lattice and there would be no free electrons. It means that the valance band is completely filled and the conduction band is completely empty. Therefore, the probability of finding allowed energy status in the valance band and conduction band is fifty-fifty. In other words, every electron leaving the valance band is allowed and energy state in the conduction band. Thus the fermi level in case of intrinsic semiconductor lies in the middle of the forbidden gap.



Context of Fermi level for a semiconductor

- Fermi level of extrinsic semiconductor
  - In case of n-type semiconductor, the fermi level is shifted up towards the conduction band.
  - The energy level formed by donor impurities is called donor energy level.



- The energy level formed by the acceptor impurities in p-type semiconductor is called acceptor energy level.
- This acceptor energy level will be formed near the valance band.